

coding technique called orthogonal frequency-division multiplexing (OFDM) that splits that radio signal into several sub-signals before they reach a receiver. This greatly reduces interference. It has an approximate range of 35 m.

- ▶ **802.11b:** Released with 802.11a, 802.11b is the slowest and the least expensive standard. 802.11b transmits in the 2.4 GHz frequency band of the radio spectrum. It can handle up to 11 megabits of data per second. Its low cost made it popular but its popularity has reduced since newer standards offer better speeds at lower prices.
- ▶ **802.11g:** 802.11g was released in June 2003 and transmits at 2.4 GHz like 802.11b, but at a higher speed of 54 megabits of data per second. 802.11g is faster because it uses the same OFDM coding as 802.11a.
- ▶ **802.11n:** This is the latest standard and was released in October 2009. It transmits at 2.4 GHz using OFDM technique and offers speeds up to 150 megabits per second.

Wi-Fi has undergone many overhauls because of security concerns. Wired Equivalent Privacy (WEP) encryption was designed to protect against casual snooping but it's no longer considered secure. Because of WEP's weakness the Wi-Fi Alliance approved Wi-Fi Protected Access (WPA). Though more secure than WEP, WPA has known vulnerabilities. The more secure WPA2 using Advanced Encryption Standard was introduced in 2004 and is supported by most new Wi-Fi devices.

RFID

Radio-frequency identification (RFID) is a technology that uses radio waves to transfer data from an electronic tag, called RFID tag or label, to a receiver within the range of a few metres. These RFID tags are small and can be attached to objects to identify and track them.

This technology has been around since 1970 but it was too expensive to be used on a large scale. These tags were expensive because they were inductively coupled RFID tags which had a complex system of metal coils, antennae and glass. Then came the capacitively coupled tags which could be mass manufactured at a significantly lower cost. They used conductive carbon ink instead of metal coils to transmit data. There has also been a considerable decrease in the size of RFID. There are RFID chips as small as 0.05 x 0.05 millimeters.

Every RFID tag works in the same way:

1. Data is stored in an RFID microchip